

Understanding the Opioid Crisis: An Analysis of Overdose Trends Across States, Demographics, and Drug Types

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Abstract. The opioid epidemic continues to be a major public health challenge in the United States, resulting in substantial increases in overdose mortality over the past two decades. This project analyzes opioid overdose death trends using multiple publicly available datasets obtained from the CDC WONDER Multiple Cause of Death database for the years 1999 through 2020. After integrating and preprocessing the datasets, exploratory data analysis was performed using line charts, choropleth maps, heatmaps, and demographic visualizations to examine trends over time, across geographic regions, and among demographic groups. Predictive modeling was then conducted using Linear Regression, Random Forest, and Gradient Boosting to estimate future opioid overdose death rates. The analysis revealed a substantial increase in overdose mortality after 2013, with synthetic opioids emerging as the primary contributor during the later years of the study period. Geographic and demographic analyses also identified meaningful differences across states, age groups, sex, and race or ethnicity. Among the predictive models evaluated, Linear Regression achieved the strongest overall performance, indicating that historical mortality patterns can provide useful information for estimating future overdose rates. Overall, this project demonstrates how data analytics can be applied to better understand complex public health trends and support future research and public health decision making.

Keywords: Opioid overdose · CDC WONDER · public health · mortality trends · predictive analytics

1 Introduction

The opioid crisis remains one of the most significant public health challenges in the United States [2]. This project explores opioid overdose trends across states, demographic groups, and opioid drug types. Understanding overdose patterns through data analytics has the potential to support evidence-based public health decisions and improve resource allocation[6].

The topic was selected because of its real-world impact and its relevance to the author's background in biology and current work in a toxicology laboratory. Data for this project was collected from the CDC WONDER Multiple Cause of Death (1999-2020) database.

1.1 Goals of the Project

The primary objective of this project is to identify demographic, geographic, and drug-specific factors associated with opioid overdose mortality in the United States.

Specifically, this project aims to:

- Analyze opioid overdose trends over time.
- Compare overdose rates across demographic groups and geographic regions.
- Explore differences in overdose mortality by opioid drug type, including heroin, prescription opioids, methadone, synthetic opioids, and other narcotics.
- Develop predictive or statistical models, if appropriate, to identify factors associated with higher overdose rates.
- Communicate findings through data visualizations and statistical analysis.

Although this project will not solve the opioid crisis, it demonstrates how data analytics can support public health research, identify vulnerable populations, and inform data-driven decision making.

2 Background / Literature Review

The opioid epidemic has affected millions of individuals across the United States and continues to be a major public health concern[3]. Previous studies have used statistical analysis and machine learning techniques to better understand overdose patterns and identify populations at increased risk [5]. This project builds upon previous public health research by analyzing publicly available CDC WONDER mortality data and applying data analytics techniques to examine opioid overdose trends across geographic areas, demographic groups, and opioid drug types.

3 Dataset Description

The primary data source for this project is the CDC WONDER Multiple Cause of Death (1999-2020) database[4]. Rather than relying on a single dataset, an analytical dataset was created by combining the results of multiple queries. Four primary datasets were extracted to examine opioid overdose mortality from different perspectives: Opioid Overdose Deaths by State and Year, Opioid Overdose Demographics by Age, Opioid Overdose Demographics by Sex, and Opioid Overdose Demographics by Race/Ethnicity. These datasets provide both geographic and demographic views of opioid overdose deaths across the United States.

The downloaded datasets were collected using an online query system as Microsoft Excel (.xls) files. Each file was converted to UTF-8 encoded CSV format before being imported into Python for cleaning and analysis. Using CSV files provided a consistent, machine-readable format that simplified merging, processing, and visualization throughout the project.

Data collection was performed by submitting multiple customized queries through the CDC WONDER interface. Each query grouped mortality records by different variables and filtered records using ICD-10 codes associated with opioid overdose deaths. In addition to the primary state-by-year dataset, five supplemental datasets were collected to further classify opioid overdose deaths by specific drug type: Heroin, Prescription Opioids, Methadone, Synthetic Opioids, and Other Narcotics. These supplemental datasets were collected to support analysis of mortality trends by specific opioid type. The data cleaning and integration process is described in Section 6.

The primary variables used throughout this project include state, year, population, total opioid overdose deaths, opioid overdose death rate, heroin overdose deaths, prescription opioid overdose deaths, methadone overdose deaths, synthetic opioid overdose deaths, other narcotic overdose deaths, age group, sex, and race/ethnicity. These attributes provide the foundation for analyzing geographic trends, demographic differences, and changes in opioid overdose mortality over time.

4 Project Methodology

The project will follow a structured data analytics workflow beginning with problem definition and ending with interpretation of results and project reporting. Figure 1 illustrates the implementation process.

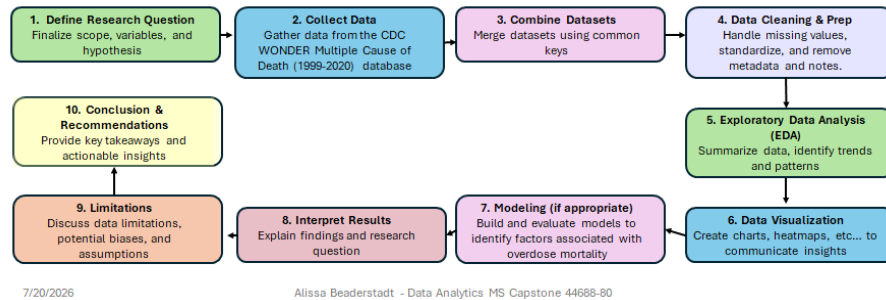


Fig. 1. Capstone Project Methodology

5 Approach and Scope

This project focuses on unintentional opioid overdose mortality in the United States from 1999 through 2020 using data from the CDC WONDER Multiple Cause of Death database. The analysis examines national and state-level trends, differences across demographic groups, and mortality involving specific opioid drug types.

The scope of the project is limited to population-level mortality data and does not include nonfatal overdoses, individual-level clinical records, or causal analysis. Because some CDC WONDER observations are suppressed, certain analyses may also be affected by missing values. These limitations are considered when interpreting the findings.

6 Data Preparation

6.1 Data Curation Process

The CDC WONDER datasets described in Section 3 were curated and prepared for analysis. The state-by-year dataset served as the foundation for the analytical dataset, while the five drug-specific datasets were merged using common state and year identifiers. The demographic datasets were maintained separately because they summarize deaths by demographic category rather than by state-year, making them incompatible with the structure of the master analytical dataset.

The preprocessing workflow included removing metadata and notes, standardizing column names, converting variables to appropriate numeric data types, and validating record counts and merge results before saving the cleaned datasets to a processed data directory.

Figure 2 illustrates the complete data curation and preprocessing workflow.

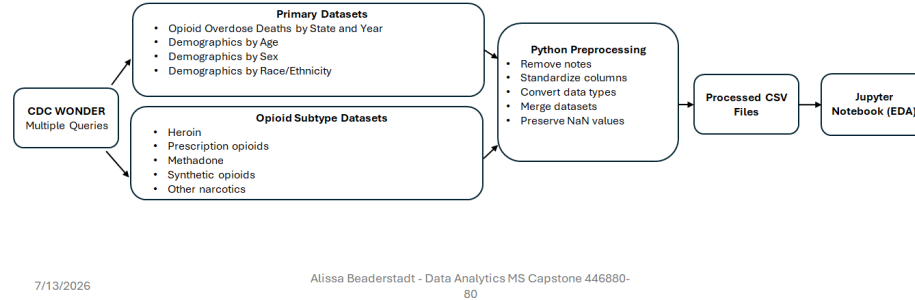


Fig. 2. Data Curation

6.2 Data Cleaning and Preprocessing

Python was used for all data preparation tasks. The pandas library was used to load, clean, merge, and export the datasets. Separate preprocessing scripts were developed for each dataset to create a reproducible data preparation workflow while keeping the processing steps organized. These scripts removed unnecessary metadata, standardized column names using snake case, converted numeric variables to appropriate data types, and merged the opioid subtype datasets into a single state-by-year analytical dataset. After cleaning and preparation,

the datasets were loaded into a Jupyter Notebook for validation and exploratory data analysis. Data quality checks included verifying data types, identifying duplicate records, inspecting missing values, and confirming the expected number of observations.

Missing values were investigated before any modifications were made to the data. Examination of the original downloaded files showed that many missing observations resulted from suppressed counts or unavailable data rather than errors introduced during preprocessing. Suppressed values were converted to NaN using the pandas `pd.to_numeric()` function and intentionally retained as missing values. These observations were not replaced with zeros because suppressed values do not indicate that zero deaths occurred. Retaining missing values preserves the integrity of the original data and prevents the introduction of misleading information into subsequent analyses.

6.3 Final Analytical Dataset

The final master analytical dataset contains 1,122 state-year observations and 15 analysis variables after removing the redundant `state_code` and `year_code` fields used only for identification. Three supporting demographic datasets were also prepared for analysis. The age dataset contains 15,155 observations, the sex dataset contains 2,244 observations, and the race/ethnicity dataset contains 4,488 observations. The size of each cleaned dataset is summarized in Table 1.

Table 1. Summary of Final Analytical Datasets

Dataset	Records	Attributes
Master	1,122	15
Age	15,155	7
Sex	2,244	7
Race/Ethnicity	4,488	7

The primary variables used throughout this analysis are defined below, and Table 2 summarizes their roles as dependent and independent variables.

- **opiod_deaths**: Total opioid overdose deaths for each state-year.
- **opiod_death_rate**: Opioid overdose deaths per 100,000 population.
- **population**: Estimated state population.
- **year**: Calendar year.
- **state**: U.S. state or District of Columbia.
- **Drug subtype variables**: Number of overdose deaths involving heroin, prescription opioids, methadone, synthetic opioids, and other narcotics, along with their corresponding death rates.

Table 2. Dependent and Independent Variables

Role	Variables
Dependent	Opioid overdose deaths (<code>opioid.deaths</code>) and opioid overdose death rate (<code>opioid.death_rate</code>), depending on the analysis performed.
Independent	State, year, population, opioid subtype variables (heroin, prescription opioids, methadone, synthetic opioids, and other narcotics), age, sex, and race/ethnicity.

7 Exploratory Data Analysis (EDA)

Exploratory data analysis (EDA) is the process of examining and summarizing data to better understand its structure, patterns, and trends before conducting more advanced analysis [7]. EDA is an essential phase of a data analysis project because it allows the analyst to become familiar with the data, identify meaningful patterns or unexpected observations, and assess data quality. Common EDA techniques include descriptive statistics, histograms, box plots, scatter plots, correlation analysis, heatmaps, and spatial analysis [8].

For this project, EDA was performed on the cleaned opioid overdose datasets described in the previous sections. The analysis used descriptive statistics, grouping and aggregation, year-over-year comparisons, and visualizations to examine trends over time and differences by geography, opioid type, and demographic characteristics. Visualizations included line charts, a U.S. choropleth map, and an age-by-year heatmap.

7.1 National Opioid Mortality Trends

The first stage of the analysis examined national opioid overdose mortality trends from 1999 to 2020. State-level deaths and population were aggregated by year to calculate annual national death rates, which were then visualized using a time-series line graph. The rate increased from approximately 2.09 deaths per 100,000 people in 1999 to 19.82 in 2020, with the trend becoming noticeably steeper in later years.

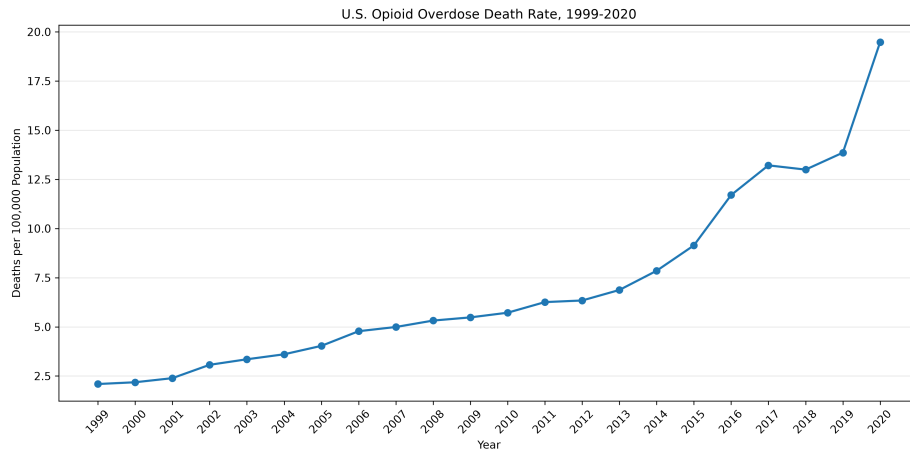


Fig. 3. National Opioid Overdose Death Rate, 1999–2020

Year-over-year changes were calculated using both the absolute and percentage change in the national death rate. Some of the largest annual increases occurred during the mid-to-late 2010s, with the largest absolute increase occurring in 2020 at approximately 3.61 deaths per 100,000 people. These comparisons helped identify periods of rapid change for further investigation.

7.2 Geographic Patterns

Geographic variation was explored using a choropleth map of state-level opioid death rates in 2020. Higher rates were concentrated in parts of the Ohio Valley, and the Northeast, while many states in the Plains and Mountain regions had comparatively lower rates.

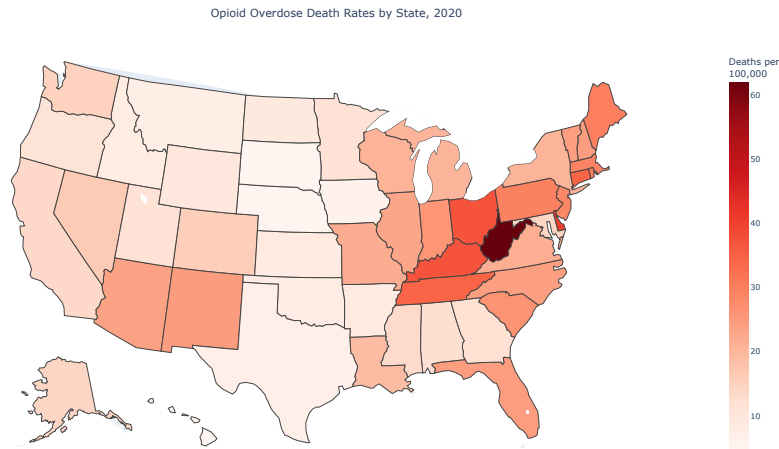


Fig. 4. U.S. Opioid Overdose Death Rates by State, 2020

7.3 Trends by Opioid Type

Time-series analysis was used to compare annual national mortality for heroin, prescription opioids, methadone, synthetic opioids, and other narcotics throughout the study period.

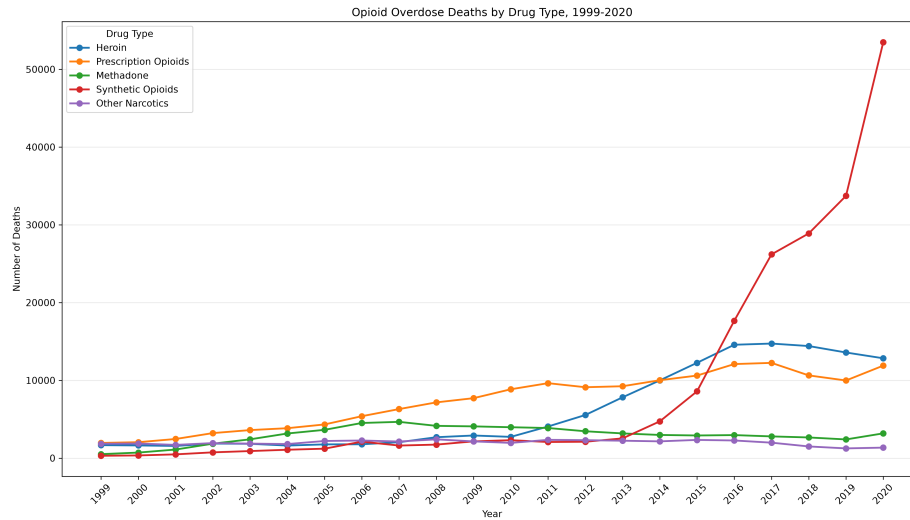


Fig. 5. Opioid Overdose Deaths by Drug Type, 1999–2020

The opioid categories showed distinct trends over time. Prescription opioid and heroin mortality increased earlier in the study period, while synthetic opioid mortality rose sharply in later years. Methadone mortality remained comparatively lower, and synthetic opioids showed the most pronounced increase by 2020.

7.4 Demographic Patterns

The demographic datasets were analyzed separately from the state-year master dataset to explore differences in opioid overdose mortality by age, sex, and race or ethnicity.

An age-by-year heatmap showed that mortality patterns changed substantially over the study period. Prior to 2013, opioid overdose death rates remained relatively low and were concentrated among adults in their late 30s through early 50s. After 2013, death rates increased rapidly across many adult age groups, with the greatest concentration shifting toward individuals approximately 25 to 40 years of age. This change mirrors the sharp increase in synthetic opioid deaths observed during the same period, which suggests that the later phase of the epidemic disproportionately affected younger working-age adults.

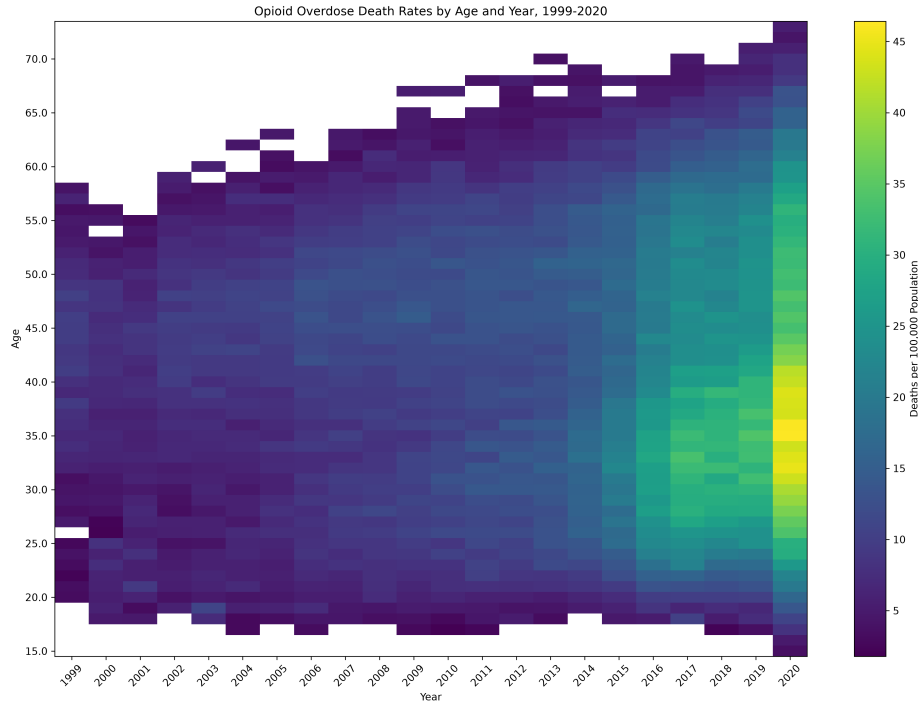


Fig. 6. Opioid Overdose Death Rates by Age and Year Heatmap

Differences by sex were examined using a time-series line chart comparing male and female opioid overdose death rates. Males consistently had higher mortality rates than females, with the gap widening in later years.

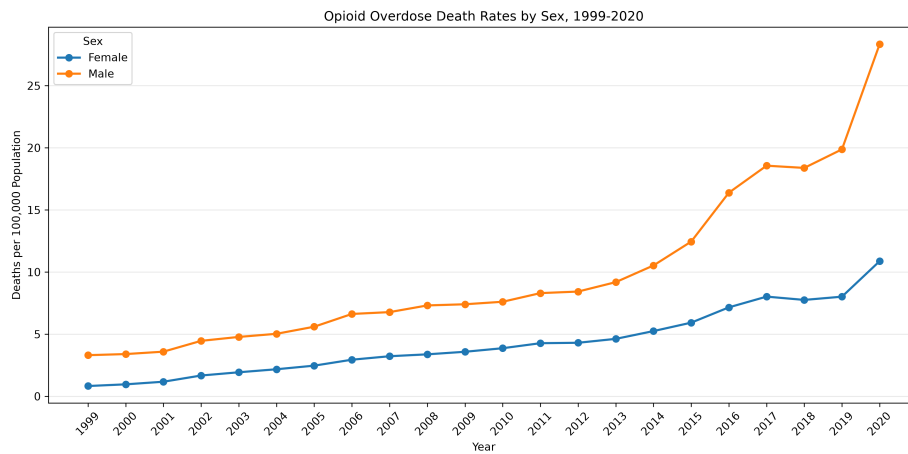


Fig. 7. Opioid Overdose Death Rates by Sex, 1999–2020

Finally, a multi-line time-series chart was used to compare opioid overdose mortality across racial and ethnic groups. White and American Indian or Alaska Native populations had elevated rates during portions of the earlier study period, while rates among Black or African American populations surpassed those groups in later years.

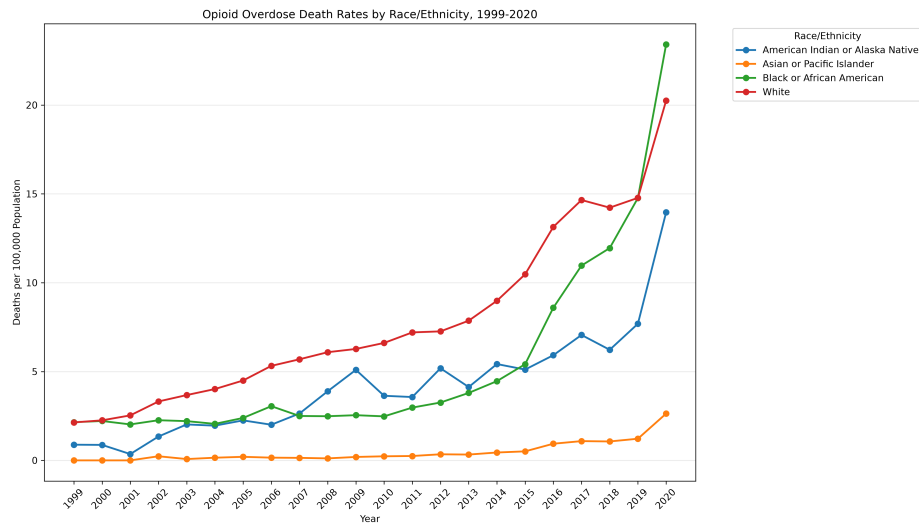


Fig. 8. Opioid Overdose Death Rates by Race/Ethnicity, 1999–2020

7.5 Key Learnings from Exploratory Analysis

The EDA phase provided a clearer understanding of how opioid overdose mortality changed over time and varied by geography, opioid type, and demographic group. It also highlighted patterns for closer examination, particularly the sharp increase in synthetic opioid mortality during the later years of the study period.

Overall, this phase helped connect the different parts of the analysis and provided direction for the next stage of the project. These findings will be examined more closely and interpreted in the Results and Analysis section.

8 Predictive Analysis

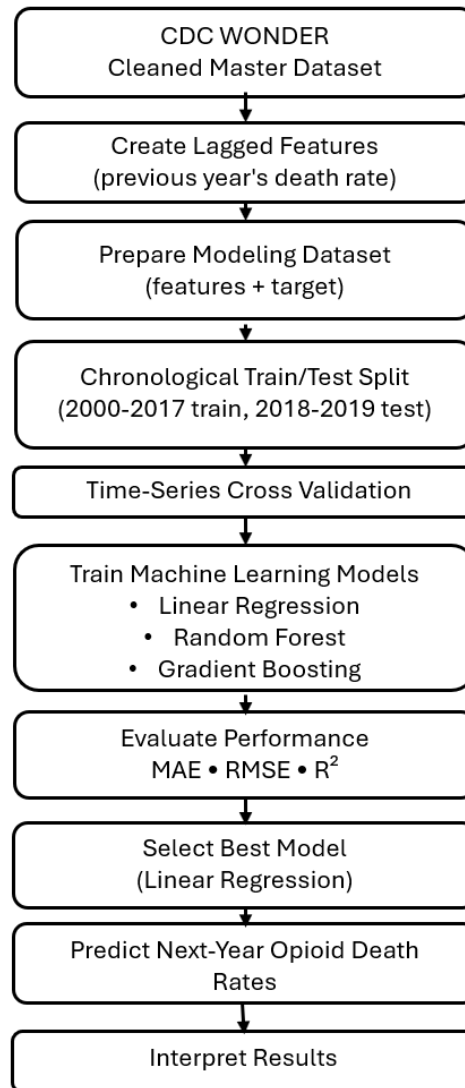
The predictive analysis phase extended the exploratory analysis by developing models to predict each state’s opioid overdose death rate for the following year. The cleaned state-by-year master dataset was used as the foundation for this phase. The goal was to determine whether recent opioid mortality patterns could provide reliable estimates of next-year opioid overdose mortality at the state level.

8.1 Predictive Pipeline

The predictive pipeline began by loading the cleaned state-by-year dataset and sorting observations by state and year. Time-based features were then created by generating a previous-year opioid death rate and a next-year prediction target

using lagged values within each state. Only consecutive yearly observations were retained to prevent gaps in the time series.

The final model used the observation year, current opioid death rate, and previous-year opioid death rate to predict the following year's opioid death rate. Records missing any required values were removed, resulting in 927 complete observations (82.6% of the original dataset).



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Fig. 9. Predictive Modeling Workflow

8.2 Machine Learning Algorithms

Three regression algorithms were evaluated: Linear Regression, Random Forest Regression, and Gradient Boosting Regression. Linear Regression served as an interpretable baseline model, while Random Forest and Gradient Boosting were included to evaluate whether nonlinear ensemble methods could improve

predictive performance. All three models were trained using the same predictor variables, training data, and evaluation metrics to ensure a consistent comparison.

8.3 Training, Validation, and Testing

Because the objective was to predict future opioid mortality, the data were split chronologically rather than randomly. Predictor years 2000 through 2017 were used for training (825 observations), while predictor years 2018 and 2019 were reserved as an independent test set (102 observations). Because the target variable represented the following year’s opioid death rate, the test set was used to predict outcomes for 2019 and 2020. This chronological split better reflects a real-world forecasting scenario and prevents future information from being used during model training.

Model validation was performed using four time-based validation folds within the training data, with 2014 through 2017 serving as validation years. Random Forest and Gradient Boosting were tuned using `GridSearchCV` and these validation folds, while Linear Regression required no hyperparameter tuning. The final test set remained completely separate and was used only for the final evaluation of model performance.

8.4 Implementation and Evaluation

The predictive analysis was implemented in Python using `pandas` for data preparation, `scikit-learn` for model development and evaluation, and `Matplotlib` for visualization. Random Forest and Gradient Boosting were tuned using `GridSearchCV` with negative mean squared error as the optimization criterion.

Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). Lower MAE and RMSE indicate better predictive accuracy, while higher R^2 values indicate better model fit. The source code and predictive modeling notebook are available in the project’s GitHub repository [1].

8.5 Predictive Modeling Results

Table 3 summarizes the test-set performance of the three predictive models.

Table 3. Test-set performance of the predictive models

Model	MAE	RMSE	R^2
Linear Regression	2.91	4.49	0.826
Gradient Boosting	3.18	4.91	0.792
Random Forest	3.21	4.93	0.790

Linear Regression achieved the best overall performance, producing the lowest MAE (2.91), lowest RMSE (4.49), and highest R^2 (0.826). Although the ensemble models performed well, neither outperformed the simpler linear model. These results confirmed that Linear Regression was the most appropriate model for the final predictive application.

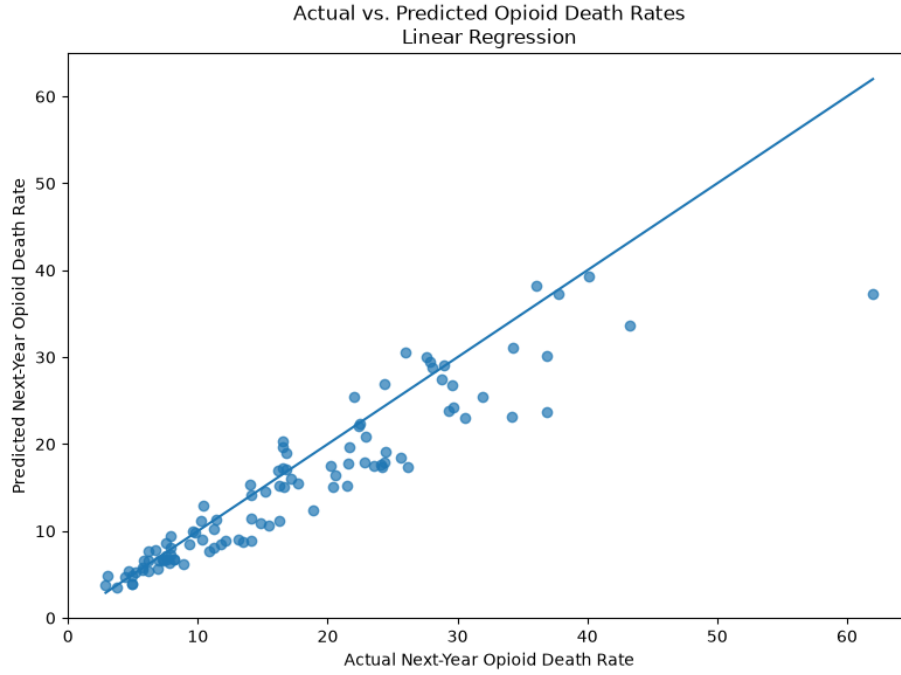


Fig. 10. Actual versus predicted next-year opioid death rates for the Linear Regression model. Points closer to the reference line ($y = x$) indicate more accurate predictions.

Figure 10 shows that the predicted values closely follow the observed opioid death rates, indicating that recent opioid mortality patterns provide useful information for predicting the following year's opioid death rate.

Prediction accuracy differed by target year. The average absolute error was 1.36 for 2019 compared with 4.46 for 2020. Many 2020 observations were under-predicted because the sharp increase in opioid overdose mortality that year was not well represented in the historical training data. This highlights an important limitation of predictive models based on historical trends: unexpected changes are difficult to anticipate when similar conditions have not previously occurred.

9 Results and Analysis

The exploratory data analysis and predictive modeling phases provided several important insights into opioid overdose mortality trends in the United States. Line charts, choropleth maps, heatmaps, and predictive models were used to examine historical trends and evaluate the ability to estimate future opioid overdose death rates.

The national trend analysis (Figure 3) showed that opioid overdose death rates increased substantially between 1999 and 2020, with particularly rapid growth occurring after approximately 2013. Geographic analysis (Figure 4) demonstrated that opioid overdose mortality varied considerably across states, with consistently higher rates observed throughout Appalachia, the Ohio Valley, and portions of the Northeast. Together, these visualizations illustrate that the opioid epidemic intensified over time while also affecting some regions much more severely than others.

The drug-specific time-series analysis (Figure 5) revealed important differences among opioid categories. Prescription opioid deaths increased during the early years of the study period before leveling off, while heroin deaths rose during the middle of the study period. In contrast, synthetic opioid deaths increased dramatically after 2013 and became the dominant contributor to opioid overdose mortality by 2020. These findings suggest that synthetic opioids became the primary driver of overdose mortality during the later years of the study period.

Demographic analyses (Figures 6–8) identified meaningful differences across age, sex, and race or ethnicity. The age heatmap (Figure 6) showed that mortality shifted from primarily affecting adults in their late thirties through early fifties before 2013 to adults approximately 25 to 40 years of age during the later years of the study period. The sex comparison (Figure 7) demonstrated that males consistently experienced higher opioid overdose death rates than females, with the gap becoming more pronounced after 2013. The race and ethnicity trends (Figure 8) showed that although White and American Indian or Alaska Native populations experienced elevated mortality during portions of the earlier study period, Black or African American populations experienced substantial increases in mortality during the later years.

Predictive modeling results (Table 3) demonstrated that historical opioid overdose data can be used to estimate future opioid overdose death rates with reasonable accuracy. Among the models evaluated, Linear Regression produced the strongest overall performance, achieving an R^2 value of approximately 0.83 while also producing the lowest prediction errors based on MAE and RMSE. Although Random Forest and Gradient Boosting also produced reasonable predictions, neither outperformed the simpler linear model. The actual versus predicted values shown in Figure 10 further illustrate this performance, with most predictions closely following the reference line. Larger prediction errors occurred for some 2020 observations, suggesting that historical trend-based models may have difficulty accurately predicting sudden, large-scale shifts that were not represented in the training data.

Overall, the results demonstrate that opioid overdose mortality has increased substantially over the past two decades while also evolving geographically, demographically, and by opioid type. The predictive analysis further suggests that historical mortality patterns can provide meaningful information for estimating future overdose rates, potentially supporting public health planning and resource allocation.

10 Limitations

Several limitations should be considered when interpreting the results of this study. First, the analysis relies exclusively on publicly available CDC WONDER mortality data, which are subject to reporting practices, death certificate accuracy, and suppressed values for small populations. These factors may result in missing data for certain states, years, or demographic groups.

Second, this project focuses on identifying associations and historical trends rather than establishing causal relationships. Although changes in mortality patterns were observed over time, the analyses cannot determine the specific causes of those changes.

Additionally, the predictive models were developed using historical mortality data and a limited set of variables available within the analytical dataset. Important external factors such as socioeconomic conditions, healthcare access, substance use treatment availability, policy changes, and fentanyl market dynamics were not incorporated into the predictive models. Including these variables may improve predictive performance in future research.

Finally, the analysis examines data through 2020 and does not capture more recent developments in the opioid epidemic.

11 Conclusion

This project demonstrates how publicly available health data can be integrated, cleaned, visualized, and analyzed to better understand opioid overdose mortality trends in the United States. By combining multiple CDC WONDER datasets, a comprehensive analytical dataset was developed that allowed geographic, demographic, and drug-specific trends to be examined over more than two decades.

The exploratory analysis revealed substantial increases in opioid overdose mortality over time, particularly after 2013, while highlighting important differences across states, opioid types, age groups, sex, and race or ethnicity. Predictive modeling further demonstrated that historical mortality patterns can be used to estimate future opioid overdose death rates with reasonable accuracy, with Linear Regression providing the strongest overall performance among the models evaluated.

Overall, the findings from this project illustrate the value of data analytics for understanding complex public health challenges. Although additional variables could improve future predictive models, the analytical workflow developed in

this project provides a strong foundation for continued research and may support future public health surveillance, decision making, and resource planning.

12 Future Work

Future work could expand this analysis by incorporating additional variables such as socioeconomic indicators, healthcare access, prescription rates, and state-level policy changes to better explain differences in opioid overdose mortality. More advanced forecasting approaches and newer mortality data could also be incorporated to evaluate whether predictive performance improves as the epidemic continues to evolve.

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